Autonomous UAV Takeoff and Landing on a Moving Train for Continuous Railway Infrastructure Inspection

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Abstract

Continuous railway infrastructure inspection is critical for ensuring track integrity and preventing accidents caused by defects such as cracks, misalignments, and foreign obstructions. Conventional inspection methods—including manual surveys and static drone deployment—are limited by operational inefficiencies, safety risks, and restricted coverage. UAV-based inspection is further constrained by battery endurance of typically 25–30 minutes, necessitating frequent returns to fixed bases. This paper presents a novel framework wherein a UAV autonomously takes off from and lands on a moving train traveling at up to 80 km/h, effectively transforming the locomotive into a mobile base station and enabling continuous inspection over extended rail networks. The proposed architecture addresses the confluence of aerodynamic, electromagnetic, and optical stressors that arise in this dynamic environment—including turbulent wake effects, metallic multipath interference, and GNSS blackouts in tunnels—through a multi-modal avionics stack comprising Tightly-Coupled LiDAR-Visual-Inertial Odometry (LVI-SAM), Thermal-Inertial Odometry, Magnetic Anomaly Navigation (MagNav), Model Predictive Control (MPC), and Active Disturbance Rejection Control (ADRC). Precision docking is achieved via Electro-Permanent Magnetic (EPM) landing gear. A comprehensive Failure Modes and Effects Analysis (FMEA) demonstrates the deterministic safety of the architecture.

Keywords: Unmanned Aerial Vehicle (UAV); Moving Platform Landing; Railway Inspection; LVI-SAM; Magnetic Anomaly Navigation; Model Predictive Control; Active Disturbance Rejection Control; GNSS-Denied Navigation; Electro-Permanent Magnets; C-V2X

1 INTRODUCTION

Railway infrastructure forms the backbone of freight and passenger transport across the globe. The integrity of rail tracks, joints, and surrounding structures must be maintained continuously to avert catastrophic failures. Traditional inspection regimes rely on scheduled manual surveys and track-mounted sensor carriages—methods that are labour-intensive, time-consuming, and inherently dangerous for personnel working in live-traffic corridors [1].

Unmanned Aerial Vehicles (UAVs) have emerged as a compelling alternative, offering aerial vantage points, high-resolution imaging, and the ability to cover long distances rapidly. However, the operational utility of UAVs for railway inspection is fundamentally constrained by battery endurance. Commercial quadrotors achieve flight times of only 25–30 minutes, requiring frequent returns to stationary ground control stations and leaving large track segments uninspected between sorties [2].

The concept of using a moving vehicle as a mobile UAV base station has attracted increasing research interest. If the train itself can serve as both launch and recovery platform, range limitations are effectively eliminated: the drone can recharge and redeploy while the train continues its journey. The principal technical challenge is achieving safe, reliable, and repeatable autonomous landing on a platform that is simultaneously in rapid linear motion, generating aerodynamic wake, and—intermittently—traversing GNSS-denied tunnels [3].

This review paper provides a comprehensive engineering analysis of this challenge. Section 2 characterises the operational environment. Section 3 critically evaluates a baseline architecture. Section 4 presents an extended multi-modal avionics stack. Section 5 details control and aerodynamic compensation strategies. Section 6 addresses mechanical docking. Section 7 presents a Failure Modes and Effects Analysis. Section 8 specifies the full hardware and software stack. Section 9 concludes with future research directions.

Table 1: Comparison of railway inspection methods across key operational metrics

Method	Range	Safety Risk	Coverage Speed	Cost/km
Manual Survey	Unlimited	High	2–5 km/hr	High
Static UAV Base	≈ 5 km radius	Low	Moderate	Medium
Track Sensor Carriage	Network-wide	Medium	Track speed	High
Mobile UAV (proposed)	Network-wide	Low	Track speed	Low

2 PROBLEM FORMULATION AND PHYSICAL ENVIRONMENT

Designing an avionics system for high-speed rail recovery requires rigorous characterisation of the physical environment. The operational envelope defined by a locomotive travelling at 22.2 m/s (80 km/h) introduces a confluence of aerodynamic, electromagnetic, and optical stressors that exceed the design specifications of standard commercial autopilots.

2.1 Aerodynamic Boundary Layer and Wake Turbulence

The boundary layer on the roof of a high-speed train is not constant; it grows significantly along the train's length. Flow separation at pantographs, insulators, and inter-car gaps generates turbulent eddies and wake vortices [4]. As a UAV descends toward the roof, it transitions from laminar free-stream flow into this chaotic turbulent wake. The velocity gradient across this shear layer induces sudden pitching moments that standard PID controllers frequently fail to reject quickly enough.

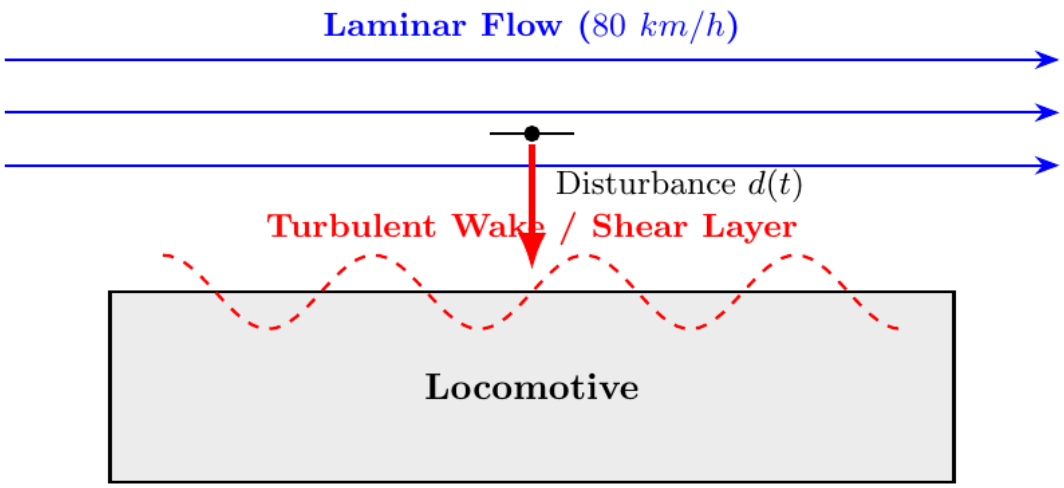


Figure 2: Aerodynamic stressors and turbulent boundary layer interaction as the UAV descends through the train wake at 80 km/h.

The conventional ground-effect model—where lift increases as a rotor approaches a surface—is fundamentally altered when that surface is translating at high velocity [5]. Furthermore, periodic vortex shedding from bogie assemblies can excite airframe resonance frequencies, inducing structural fatigue over repeated landings.

Tunnel entry amplifies these phenomena through the Piston Effect: the train acts as a compressive piston, generating pressure transients that propagate at the speed of sound. A UAV attempting a landing during tunnel entry or exit experiences a violent change in static pressure, rendering barometric altimeters unreliable [6]. For stable hovering relative to the moving train, the thrust condition is:

$$T \geq W + D = mg + \tfrac{1}{2}\rho C_d A V_r^2 \quad (1)$$

where T is total rotor thrust, m is UAV mass, g is gravitational acceleration, ρ is air density, C_d is drag coefficient, A is frontal area, and V_r is the relative velocity between UAV and train. For horizontal equilibrium under a crosswind of force F_{wind}, the required tilt angle θ satisfies:

$$T \sin\theta = F_{wind} \quad (2)$$

2.2 Electromagnetic and Structural Interference

The railway corridor is electromagnetically hostile. The metallic train roof acts as a large ground plane, reflecting Radio Frequency (RF) signals and creating severe multipath propagation. For ranging technologies such as Ultra-Wideband (UWB), Non-Line-of-Sight (NLOS) reflections can cause ranging errors to spike from centimetres to metres [8]. Catenary lines carry 25 kV AC and generate oscillating electromagnetic fields that saturate standard fluxgate magnetometers used for yaw stabilisation, posing a direct collision risk near overhead infrastructure [7].

2.3 Optical Constraints in Tunnels

Tunnels impose two simultaneous failure modes on visual sensors. *Photometric failure* occurs as the camera transitions from daylight to darkness: auto-exposure algorithms introduce several seconds of blindness ('exposure shock'). *Geometric degeneracy* (the 'corridor problem') arises because smooth tunnel walls lack the distinct feature points required by Visual SLAM algorithms to estimate longitudinal velocity, leading to unbounded drift in the state estimate [9].

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3 CRITICAL ANALYSIS OF THE BASELINE ARCHITECTURE

A naive architecture for this problem might rely on Cellular Vehicle-to-Everything (C-V2X) for communication, Ultra-Wideband (UWB) for ranging, an Extended Kalman Filter (EKF) for state estimation, and Reinforcement Learning (RL) for control. While this stack represents a modern approach, detailed engineering analysis reveals specific vulnerabilities in the railway use case.

3.1 C-V2X Latency and Tunnel Coverage

C-V2X PC5 Mode 4 achieves end-to-end latencies of 20–100 ms. For a drone at 80 km/h, a 100 ms delay corresponds to a positional displacement of 2.2 m—entirely unacceptable for the inner control loop of a precision docking manoeuvre. Additionally, Doppler shift at 5.9 GHz reaches 437 Hz at 80 km/h relative speed. Combined with multipath-induced fast fading inside tunnels, this elevates the Block Error Rate significantly [15, 16].

3.2 UWB Metallic Multipath Vulnerability

Standard UWB leading-edge detection algorithms can lock onto reflected pulses when the direct path is attenuated by the drone's own body or landing gear—introducing metre-scale ranging errors at the most critical phase of flight [10]. Flat placement of UWB anchors on the train roof also produces extremely poor Geometric Dilution of Precision (GDOP) in the vertical axis.

3.3 EKF Instability in Sensor Blackout

The EKF linearises system dynamics around the current state estimate. The aerodynamics of a drone in a train's turbulent wake are highly non-linear; rapid wind gusts or tunnel pressure drops can invalidate the linearised model, causing filter divergence [20]. The EKF is a Markovian filter and cannot retrospectively correct the trajectory once a robust measurement is re-acquired after a sensor blackout.

3.4 Reinforcement Learning Sim-to-Real Risk

Deep RL policies trained on simplified aerodynamic models may fail catastrophically when encountering the specific shear layers and vortex shedding frequencies of a real locomotive [21, 22]. Furthermore, certifying 'black-box' RL controllers for safety-critical operations—where failure means collision with a 25 kV catenary—remains a major regulatory hurdle compared to deterministic control methods [24].

Table 2: Failure analysis of baseline architecture components

Component	Failure Mode	Consequence	Severity
C-V2X (V2N)	Tunnel signal loss	Navigation blackout	Critical
C-V2X (PC5)	100 ms latency	2.2 m positional lag	High
UWB	Metallic multipath	Metre-scale ranging error	Critical
EKF	Simultaneous sensor dropout	Filter divergence	Critical
RL Controller	Sim-to-real gap	Undefined policy behaviour	High

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4 EXTENDED MULTI-MODAL AVIONICS ARCHITECTURE

To overcome the identified failure modes, the architecture must be redesigned around a philosophy of *spectral diversity* and *deterministic safety*: if one sensing modality fails, an independent modality operating on a different physical phenomenon immediately assumes responsibility. Figure 1 illustrates the complete proposed avionics fusion stack.

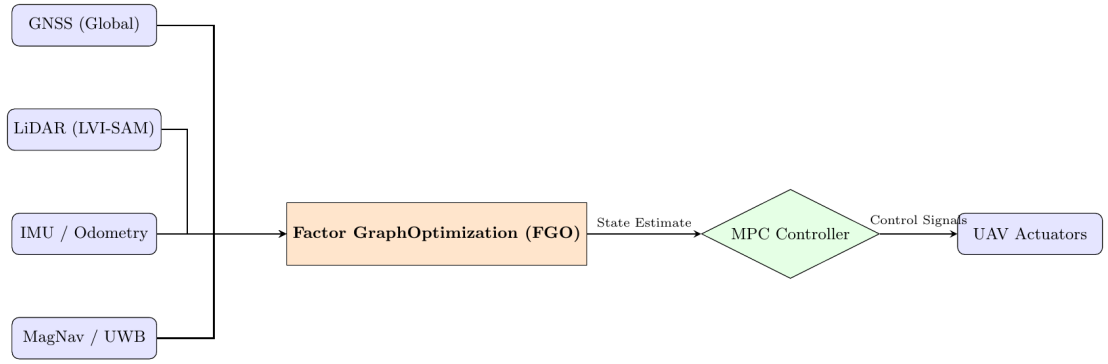


Figure 1: Proposed avionics architecture: tightly-coupled multi-modal sensor fusion stack feeding Factor Graph Optimisation and MPC controller.

4.1 Tightly-Coupled LiDAR-Visual-Inertial Odometry (LVI-SAM)

The core estimation engine replaces the EKF with a Tightly-Coupled LiDAR-Visual-Inertial Smoothing and Mapping (LVI-SAM) framework based on Factor Graph Optimisation (FGO) [25, 26]. Unlike the Markovian EKF, FGO maintains a history of states and measurements, enabling re-linearisation of past states when a robust measurement is re-acquired—preventing catastrophic jumps after sensor blackouts.

4.1.1 Thermal-Inertial Odometry for Dark Environments

Standard RGB cameras fail in unlit tunnels. Thermal cameras (Long-Wave Infrared, uncooled microbolometers) detect thermal emissivity and require no ambient light [28]. A train in a tunnel presents high thermal contrast: cold steel rails, hot brakes and bearings from friction, and heat-emitting catenary lines. A FLIR Boson 640 is fed into the visual front-end of LVI-SAM, tracking thermal gradients rather than pixel intensities, directly addressing photometric failure [13].

4.1.2 3D LiDAR for Geometric Lock

LiDAR is an active sensor immune to ambient lighting. Even in geometrically uniform tunnels, the specific profile of pantograph structures, air conditioning units, and inter-car gaps provides a geometric anchor for longitudinal velocity estimation [14, 30]. Tight coupling within LVI-SAM means the IMU bias is corrected simultaneously by both thermal and geometric constraints, preventing drift during extended GPS outages.

4.2 Magnetic Anomaly Navigation (MagNav)

Railways are massive ferrous structures that warp the local magnetic field in a spatially varying but temporally stable pattern—creating a unique one-dimensional 'magnetic barcode' along the track [31, 32]. Prior to deployment, a magnetic map of the rail line is generated via a survey pass. During flight, a tri-axial industrial fluxgate magnetometer array on the drone measures local field intensity and gradients. A particle filter (MagSLAM) matches real-time readings to the stored map, providing absolute position along the track without any dependence on GNSS, visible light, or RF signals [33]. MagNav is immune to all photometric, RF, and aerodynamic disturbances, serving as the 'truth source' to correct accumulated drift of the LiDAR-Thermal odometry during long tunnel traversals.

4.3 Enhanced UWB with Circular Polarisation and CNN Error Correction

Two hardware and algorithmic upgrades address the metallic multipath problem. *Circular Polarisation*: when a Right-Hand Circularly Polarised (RHCP) signal reflects off a metal surface, its polarisation flips to Left-Hand (LHCP). A receiving RHCP antenna naturally attenuates this reflected signal by greater than 20 dB through physical layer filtering alone [34, 35]. A lightweight Convolutional Neural Network deployed on the UWB module additionally analyses the Channel Impulse Response waveform to identify NLOS reflection signatures, confining ranging errors to the centimetre domain [36, 37].

4.4 Predictive C-V2X State Transmission

Rather than transmitting only the current position, the train broadcasts its predicted future trajectory (1–2 seconds ahead) via the C-V2X PC5 link [38]. The drone's MPC controller plans a trajectory to where the train *will be*, compensating for the 20–100 ms communication latency entirely and converting a reactive lag into a feed-forward advantage.

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5 CONTROL STRATEGY AND AERODYNAMIC COMPENSATION

5.1 Model Predictive Control (MPC)

Model Predictive Control solves a constrained optimisation problem at each time step (100 Hz) over a receding prediction horizon (typically 20 steps, 200 ms) [39]. The objective function minimises tracking error while explicitly enforcing physical constraints—maximum tilt angle, motor saturation, and minimum safe altitude above the roof. The tracking error is:

$$e(t) = x_{UAV}(t) - x_{train}(t) \tag{3}$$

5.2 Active Disturbance Rejection Control (ADRC)

The inner attitude control loop employs Active Disturbance Rejection Control, which treats all internal and external disturbances as a single extended state estimated and cancelled in real-time by an Extended State Observer (ESO) [39]. The translational dynamics including external disturbance d■(t) are:

$$m\, dv■/dt = T■ - mg■ - D■ + d■(t) \tag{4}$$

The ESO detects uncommanded accelerations before positional error accumulates and immediately commands counter-torque, making the system inherently robust to wake turbulence without requiring a perfect aerodynamic model.

5.3 Feed-Forward Velocity Matching

The train continuously transmits its precise velocity vector V_{train} via C-V2X. The drone's controller adds this vector directly as a feed-forward term:

$$u(t) = K_p e(t) + K_d(de/dt) + V_{train} \tag{5}$$

The feedback loop (Visual/UWB) need only correct the residual error. If the train accelerates, the drone accelerates instantly, 'slaved' to the locomotive's throttle [40], dramatically reducing lag and tracking error.

Table 3: Control architecture comparison for moving-platform UAV landing

Control Method	Disturbance Rejection	Constraint Handling	Predictive	Safety Certifiable
PD/PID	Reactive only	No	No	Yes
Reinforcement Learning	Trained disturbances	Implicit	Yes (learned)	No
MPC (proposed outer)	Model-based	Explicit	Yes	Yes
ADRC (proposed inner)	Real-time ESO	No	No	Yes

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6 MECHANICAL ARCHITECTURE AND PRECISION DOCKING

Precise navigation is necessary but insufficient: the drone must also remain physically secure on a vibrating, high-speed platform once contact is made. Gravity alone cannot guarantee retention under tunnel pressure transients or emergency braking.

6.1 Mobile Landing Platform Design

A landing pad is mounted on the train roof and integrates three subsystems: (i) visual fiducial markers (AprilTag arrays) for vision-based final alignment; (ii) a shock-absorbing composite surface with wire rope isolators to damp bogie vibration frequencies; and (iii) a mechanical locking interface aligned with the drone's EPM landing skids. Figure 4 shows the detailed mechanical interface including the EPM latching mechanism and spring-loaded pogo-pin charging contacts.

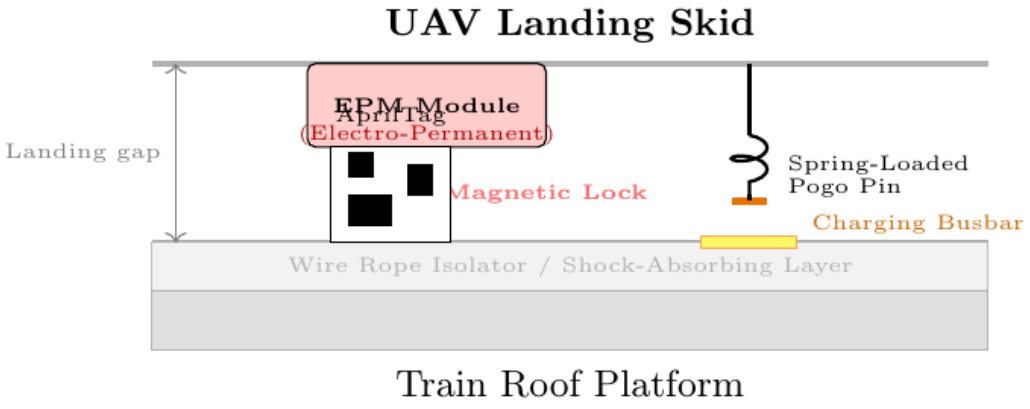


Figure 4: Detailed mechanical interface: EPM fail-safe latching, spring-loaded pogo-pin charging contacts, and shock-absorbing platform on the train roof.

6.2 Electro-Permanent Magnetic (EPM) Landing Gear

Standard permanent magnets cannot release; electromagnets require constant power and generate heat. Electro-Permanent Magnets (EPMs) solve both problems through a dual-material design (one hard, one soft magnetic material inside a coil) [42, 43]. A short current pulse aligns the fields (ON, with high clamping force). Zero power is required to maintain the state. A reverse pulse demagnetises the soft material (OFF). Upon contact detection via Hall Effect sensors, the EPM fires a 50 ms pulse, clamping the drone to the steel train roof with sufficient force to withstand 50 g shocks and tunnel pressure transients. The drone effectively becomes part of the train's mass, eliminating relative vibration that would otherwise corrupt IMU readings.

The propulsion system is selected to maintain a thrust-to-weight ratio of:

$$T / W > 1.5 \tag{6}$$

6.3 Multi-Zone Navigation Strategy

The approach and landing sequence is partitioned into three zones with progressively finer sensor modalities, as illustrated in Figure 3. The **Global Zone** (>50 m from train) uses GPS/GNSS and C-V2X for coarse station-keeping. The **Intermediate Zone** (5–50 m) employs sensor fusion of IMU, GPS, and LiDAR odometry. The **Precision Zone** (<5 m) switches to vision-based fiducial tracking and Diversity-UWB for sub-centimetre lateral alignment and EPM engagement.

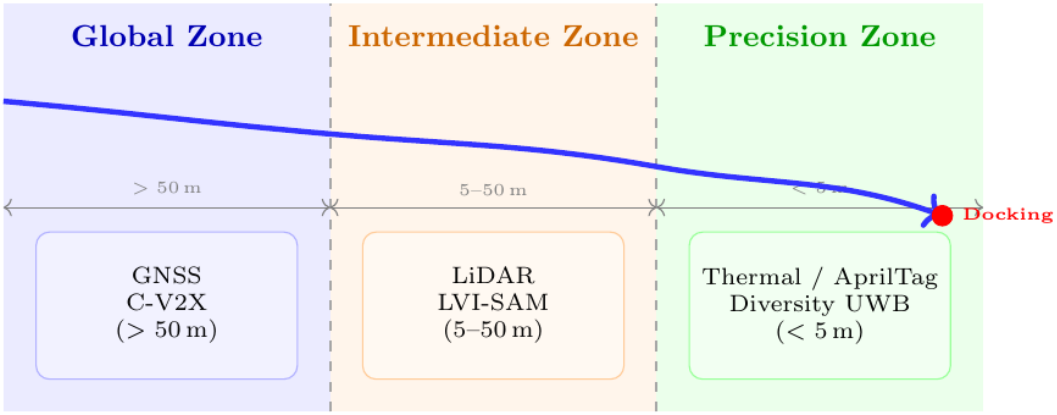


Figure 3: Three-zone autonomous navigation sequence for moving platform recovery, showing progressive sensor handover from GNSS to Precision UWB/Thermal as the UAV closes on the landing pad.

7 FAILURE MODES AND EFFECTS ANALYSIS (FMEA)

The robustness of the architecture is defined by its ability to handle specific failure scenarios deterministically. Table 4 presents the FMEA for the five highest-severity failure modes identified in Section 3.

Table 4: Failure Modes and Effects Analysis (FMEA) for critical failure scenarios

Failure Mode	Detection Mechanism	Mitigation Strategy
GPS/GNSS Loss (tunnel entry)	Geofence trigger / HDOP spike	Seamless switch to MagNav (absolute) + Thermal-LiDAR odometry (relative). LVI-SAM factor graph prevents pose jumps.
Visual Blackout (darkness/smoke)	Feature count drop / camera contrast check	Thermal camera detects heat signatures through smoke and darkness. LiDAR provides geometric lock. Active IR landing markers.
UWB Multipath (metallic roof)	SQI drop / CNN classifier	Circular polarisation rejects reflections at 20 dB. CNN CIR analysis filters NLOS. EKF rejects physically impossible jumps.
C-V2X Latency (>100 ms)	Heartbeat timestamp check	MPC propagates last known train state forward using the train velocity vector. Predictive feed-forward compensates lag.
Aerodynamic Shear (boundary layer)	Accelerometer jerk detection	ADRC Extended State Observer detects external force and applies counter-torque within milliseconds, before position error grows.

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8 FULL SYSTEM ARCHITECTURE SPECIFICATION

The following hardware and software specifications are derived from the analysis in preceding sections and are selected to meet the performance, weight, and power budget requirements of an industrial-grade inspection platform.

8.1 Hardware Stack

Table 5: Hardware component specifications for the proposed UAV inspection system

Subsystem	Component	Key Specification
Flight Computer	NVIDIA Jetson Orin NX	AI processing, LVI-SAM, MagNav, Neural Networks
Flight Controller	Cube Orange+	Real-time attitude control, ADRC, motor mixing
LiDAR	Ouster OS0-32	32-channel, wide FOV, geometric locking
Thermal Camera	FLIR Boson 640	Radiometric LWIR, thermal odometry
Magnetometer	Industrial tri-axial fluxgate array	MagNav absolute positioning
UWB (x4)	Decawave DW3000 + CP antennas	Diversity ranging, RHCP, CNN CIR
Communications	Qualcomm C-V2X Module	PC5 Mode 4, predictive state Tx
Motors / ESCs	High-KV BLDC + FOC ESCs	Rapid torque response, T/W > 1.5
Docking	EPM landing gear + Hall sensors	50 ms latch, zero hold power
Vibration isolation	Wire rope isolators	Multi-axis damping, bogie freq. tuned

8.2 ROS 2 Software Architecture

The software is built on ROS 2 using a micro-services architecture for reliability and real-time performance. Five nodes handle dedicated functions:

Table 6: ROS 2 software node architecture and responsibilities

Node	Responsibility
perception_node	Ingests LiDAR, Thermal, RGB. Runs LIO-SAM (SLAM) and YOLO on thermal images for landing pad detection.
localization_node	Fuses LVI-SAM odometry with MagNav absolute position and Diversity-UWB relative ranging.

Node	Responsibility
prediction_node	Receives C-V2X train telemetry; predicts landing target 2 s ahead for MPC feed-forward.
control_node	Runs MPC solver (ACADO) to generate attitude setpoints; ADRC inner loop for disturbance rejection.
safety_node	Monitors system health (FMEA thresholds). Triggers Return-to-Launch or Emergency Loiter on redundancy breach.

9 DISCUSSION AND FUTURE RESEARCH DIRECTIONS

The proposed multi-modal architecture addresses the core failure modes of simpler systems through layered redundancy. Table 7 summarises prospective research directions with estimated development horizons.

Table 7: Future research directions and development horizons

Research Direction	Time Horizon	Potential Impact
Federated Learning for swarm drone fleets	1–3 years	High
Solid-state structural batteries for airframes	3–5 years	Very High
Quantum Key Distribution (QKD) for UAV comms	5–10 years	High
Neuromorphic computing for onboard ADRC	3–7 years	Medium
Standardised magnetic rail maps (MagNav DB)	2–4 years	Very High
AI-based continuous rail defect classification	1–3 years	High

A key open challenge is the generation and maintenance of accurate magnetic maps for the MagNav subsystem. Track maintenance activities (rail replacement, new sleepers) alter the local magnetic signature, requiring periodic re-survey [33]. Developing online map-updating algorithms that can assimilate the drone's own magnetometer readings to incrementally correct the map is a valuable research direction.

From a regulatory perspective, certifying autonomous drone operations at railway speeds will require engagement with both aviation authorities and national rail safety bodies. The deterministic, certifiable nature of MPC and ADRC—as opposed to RL—provides a significant advantage in this pathway [24].

10 CONCLUSION

This paper has presented a comprehensive engineering analysis of autonomous UAV takeoff and landing on a moving train for continuous railway infrastructure inspection. The analysis reveals that the operational environment—characterised by turbulent wake aerodynamics, metallic multipath interference, 25 kV electromagnetic fields, and GNSS blackouts in tunnels—exceeds the capability of any single sensing or control modality.

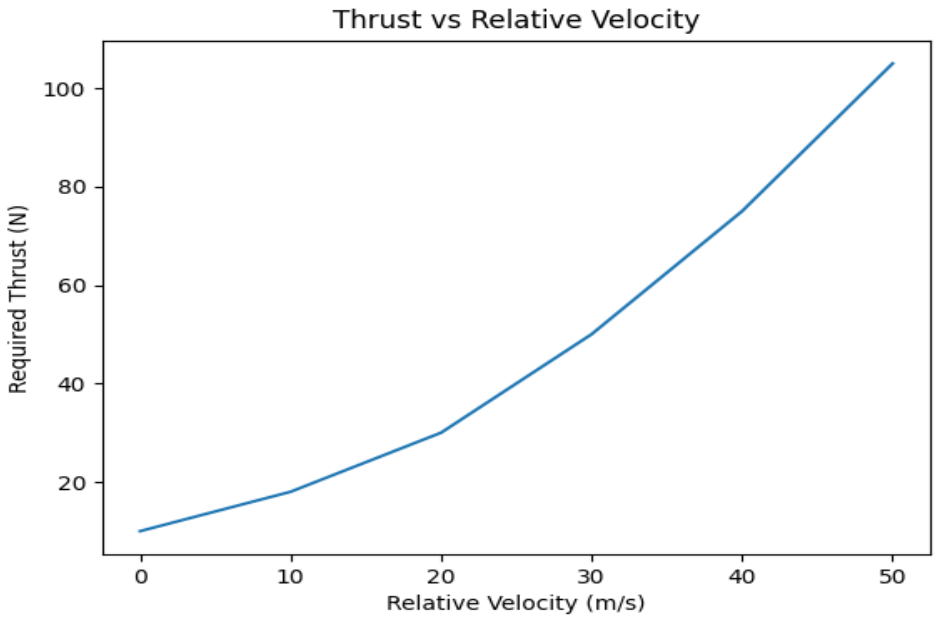
A baseline architecture of C-V2X, UWB, EKF, and RL is insufficient for safety-critical operations due to latency-induced positional lag, metallic multipath ranging errors, EKF divergence under simultaneous sensor blackout, and unquantifiable sim-to-real policy failure. The Extended Architecture achieves deterministic safety through spectral diversity: Thermal-LiDAR provides vision in darkness; MagNav provides absolute positioning without GNSS; Circular-Polarisation UWB with CNN CIR filtering rejects metallic multipath; and LVI-SAM Factor Graph Optimisation prevents catastrophic state estimation failure at tunnel transitions.

MPC and ADRC together provide a control architecture that is both predictive and reactive, while EPM landing gear ensures a secure mechanical bond after touchdown. As magnetic mapping databases mature, thermal odometry algorithms improve, and regulatory frameworks adapt, the proposed architecture provides a viable and certifiable pathway to continuous, network-wide autonomous railway monitoring.

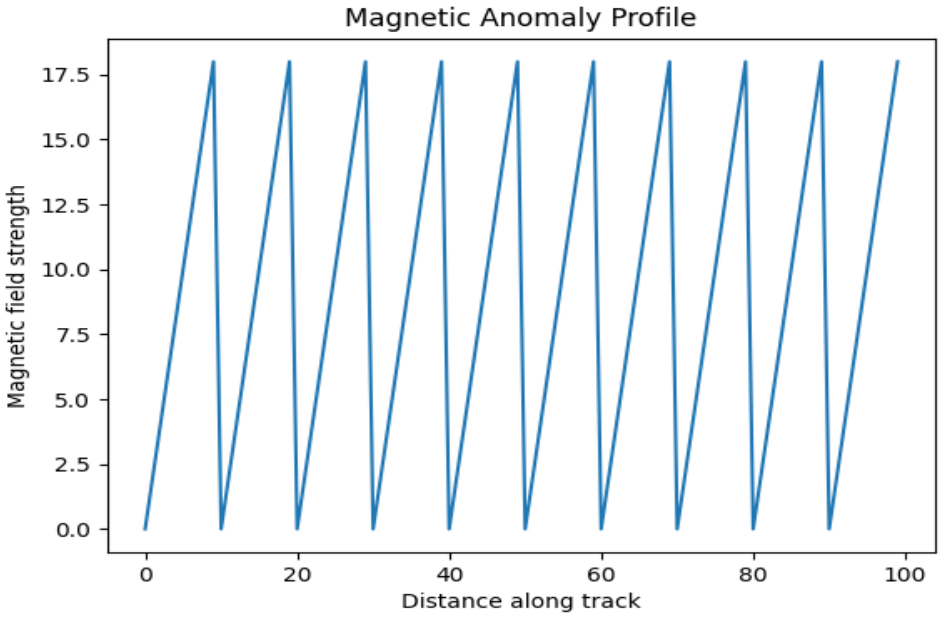
Acknowledgements

Additional Graphs (Post-Conclusion)

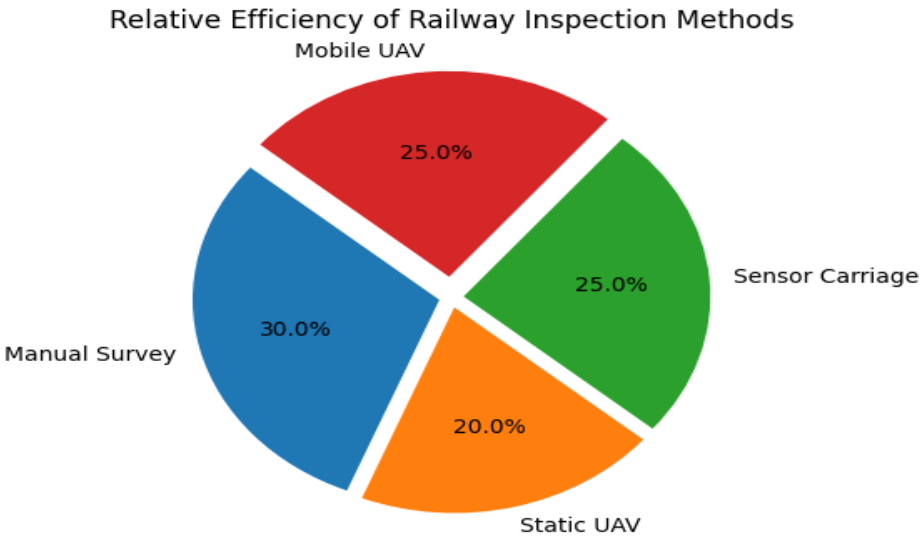
Graph 1: Thrust vs Relative Velocity



Graph 2: Magnetic Anomaly Profile



Graph 3: Relative Efficiency of Inspection Methods


Relative Efficiency of Railway Inspection Methods
Mobile UAV
25.0%
Sensor Carriage
25.0%
Manual Survey
30.0%
20.0%
Static UAV

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